



Oil prices and economic policy uncertainty: Evidence from a nonparametric panel data model

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ABSTRACT

We examine the relationship between oil prices and economic policy uncertainty in G7 countries. To do so, we employ a nonparametric panel data technique that allows the trend and coefficient functions to evolve as unknown time-varying functional forms. We also estimate country-specific and common trend functions allowing them to evolve over time. Using monthly data from G7 countries over the period 1997:01–2018:06, we find that the effect of oil prices on economic policy uncertainty is time-varying. Our results show that the estimated time-varying coefficient function of the oil price was negative in years in which increases in oil prices were driven by a surge in global aggregate demand. Further, our nonparametric local linear estimates show that the country-specific and common trend functions are increasing over time. Our findings are robust to endogeneity and alternative specifications.

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1. Introduction

We ask two important questions that are of interest to policy makers. Do changes in oil prices drive economic policy uncertainty? Does the relationship between oil prices and economic policy uncertainty vary over time? The relationship between oil prices and economic policy uncertainty has been featured in recent studies in the energy economics literature (e.g., Antonakakis et al., 2014; Degiannakis et al., 2018; Kang and Ratti, 2013a,b; Kang et al., 2017; Rehman, 2018). Given the importance of economic policy uncertainty following the Global Financial Crisis (GFC), it is of primary interest to policy makers to examine how economic policy uncertainty serves as a channel through which oil price shocks transmit to the macroeconomy. Existing studies on the relationship between oil prices and economic policy uncertainty employ parametric models with restrictive functional form assumptions. The limitation of parametric models is that the estimates may be biased and inconsistent, due to misspecification when there is no prior

knowledge of the functional form. There are no studies that use nonparametric modelling. Moreover, each of these studies apply a structural VAR framework, based on Kilian (2009), to time series data for a single country or group of countries. There are no panel studies that examine the effect of oil prices on economic policy uncertainty. Another common limitation of these studies is that long-term trends in economic policy uncertainty are ignored. It is important to account for these trends as long-term movements in economic policy uncertainty can shed light on long-term policy directions (Duca and Saving, 2018). In this study, we employ nonparametric techniques in a panel data framework to estimate long term trends in economic policy uncertainty and its time-varying relationship with oil prices.

Beginning with the seminal work of Hamilton (1983), it is well established that oil price shocks affect several macroeconomic activities including GDP, inflation and stock returns (Alquist et al., 2013; Blanchard and Riggi, 2013; Christoffersen and Pan, 2018; Cologni and Manera, 2008; Feng et al., 2017; Filis, 2010; Kilian and Park, 2009; Lippi and Nobili, 2012; Ratti and Vespignani, 2016; Tang et al., 2010). Existing studies on the macroeconomic impact of oil prices have largely focused on the business cycle channel via which oil price shocks negatively affect macroeconomic performance. For example, Hamilton (2003) shows that every recession in the US is preceded by

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oil price fluctuations. In a recent paper, [Barrero et al. \(2017\)](#) find that oil price volatility has a significant effect on short-term uncertainty. Thus, changes in oil prices may also be among the most important drivers of economic policy uncertainty. Since the work of [Bloom \(2009\)](#), a large number of papers have examined the macroeconomic effects of uncertainty (see [Bloom, 2014](#), for a survey).

[Baker et al. \(2016\)](#) construct media-based indices of economic policy uncertainty for the US and a number of other countries. They show that economic policy uncertainty has intensified following the GFC and that it remains high today in Western Europe and North America with significant and persistent adverse effects on real economic activities. A growing strand of the literature provides evidence that changes in economic policy uncertainty indices are associated with changes in a number of macroeconomic variables, such as industrial production and employment ([Baker et al., 2016](#)), the unemployment rate ([Caggiano et al., 2014, 2017](#)) and stock market returns and volatility ([Kang and Ratti, 2013a,b](#); [You et al., 2017](#)). Other strands of the literature find that economic policy uncertainty may be correlated with asset prices ([Brogaard and Detzel, 2015](#)) and bank loan growth ([Bordo et al., 2016](#)). Substantial evidence shows that policy uncertainty is negatively correlated with business cycles and that uncertainty amplifies the effect of recessions through delaying firms' investment and hiring decisions and by undermining the effectiveness of economic policies ([Bloom, 2014](#)). Therefore, investigating the factors that drive economic policy uncertainty is important in order to facilitate effective macroeconomic management.

In this study, we utilize a nonparametric panel data model with time-varying coefficients to examine the effect that changes in oil prices have on economic policy uncertainty in G7 countries. We relax the restrictive functional form assumptions in the existing literature. We also contribute to the literature by estimating the country-specific and common trend functions that are largely ignored in the literature. Another feature of our study is that we specifically address endogeneity issue in a nonparametric framework. We adopt the [Su and Ullah \(2008\)](#) procedure to address the potential problem that oil prices and economic policy uncertainty might be simultaneously determined.

Motivating the relevance of varying-coefficient nonparametric modelling, some studies, employing Markov switching models and time-varying parameter VARs, have found that the oil price – economic policy uncertainty relationship is time-varying ([Antonakakis et al., 2014](#); [Degiannakis et al., 2018](#)) and nonlinear ([Rehman, 2018](#)). The relationship between oil prices and economic policy uncertainty has become complex and unstable, meaning that parametric models may not uncover the underlying relationship between the two variables, nor the manner in which this relationship has evolved over time. While Markov switching models or time-varying parameter VARs have been used in a few studies to capture the effect of structural breaks and instability, we contend that such approaches are too restrictive to capture the complexity of the underpinning relationship (see also [Silvapulle et al., 2017](#)).

There are also advantages in applying a nonparametric framework to panel data over employing time series data for a single country or group of countries as the extant literature does. Panel data can exploit variation both over time and across cross-sectional units. Use of panel data provides additional observations which enhances degrees of freedom and allows one to address the presence of cross-sectional dependence. Thus, coefficient estimates are consistent and efficient.

Using monthly data from January 1997 to June 2018 for G7 countries, we find that the relationship between oil prices and economic policy uncertainty is time-varying. While the relationship between oil prices and economic policy uncertainty is negative for much of the period before the GFC, we find that the oil price has a strong positive effect on economic policy uncertainty in the aftermath of the GFC. Moreover, we find that the country-specific and common trend functions are increasing over time.

The remainder of this paper is organized as follows. [Section 2](#) sets out the nonparametric panel data technique. We discuss the local linear dummy variable estimator and the procedures for constructing optimal bandwidth and bootstrapping confidence intervals. [Section 3](#) discusses the data benchmark parametric estimation results and [Section 4](#) contains a discussion of the results derived from the nonparametric estimators and robustness tests. [Section 5](#) concludes the paper.

2. Nonparametric panel data model with time-varying coefficients

2.1. Local linear estimations

Following the modelling strategy in [Li et al. \(2011\)](#) and [Silvapulle et al. \(2017\)](#), our econometric model that relates the economic policy uncertainty index to the real price of oil is of the form:

$$Y_{it} = f_t + \beta_{t,1} \text{rpoil}_t + \beta_{t,2} \text{oilpvol}_t + \beta_{t,3} \text{iproduct}_t + \beta_{t,4} \text{rir}_t + \alpha_i + e_{it}. \quad (1)$$

Let $X_{it} = (\text{rpoil}_t, \text{oilpvol}_t, \text{iproduct}_t, \text{rir}_t)$ and $\beta_t = (\beta_{t,1}, \beta_{t,2}, \beta_{t,3}, \beta_{t,4})$. This model is written in matrix form as

$$Y_{it} = f_t + X_{it}^T \beta_t + \alpha_i + e_{it}, \quad \text{for } i = 1, 2, \dots, N, \quad \text{and } t = 1, 2, \dots, T, \quad (2)$$

where Y_{it} is the log of the economic policy uncertainty index, $f_t = f_t(t/T)$ are unknown country-specific trend functions, rpoil_t is the real price of crude oil, oilpvol_t denotes the oil price volatility, iproduct_t is index of industrial production and rir_t denotes real interest rate. α_i denote unobserved individual effects and e_{it} is the error term, which is stationary and independent of the regressors. $\beta_{t,j} = \beta_j(t/T)$ and $f_t = f_t(t/T)$ are vectors of time-varying coefficient and trend functions, where $f(\cdot)$ and $\beta_j(\cdot)$ are unknown smooth functions. For identification purpose, we assume that $\sum_{i=1}^N \alpha_i = 0$.

To estimate the trend functions, f_t , and the time-varying coefficients of interest, $\beta_{t,j}$, that measure the effects of oil price volatility on economic policy uncertainty in Eq. (2), we use the LLDVE method proposed by [Li et al. \(2011\)](#). To explain the LLDVE estimator in detail, we introduce the following notations.

$$\begin{aligned} \tilde{Y} &= (Y_{11}, \dots, Y_{1T}, Y_{21}, \dots, Y_{2T}, \dots, Y_{N1}, \dots, Y_{NT})^T, \\ \tilde{B}(X, \beta) &= (X_{11}^T \beta_1, \dots, X_{1T}^T \beta_T, \dots, X_{N1}^T \beta_1, \dots, X_{NT}^T \beta_T)^T, \\ \tilde{e} &= (e_{11}, \dots, e_{1T}, e_{21}, \dots, e_{2T}, \dots, e_{N1}, \dots, e_{NT})^T, \\ \tilde{f} &= \bar{I}_N \otimes (f_1, \dots, f_T)^T = \bar{I} \otimes f, \\ \alpha &= (\alpha_2, \dots, \alpha_N)^T, \\ D &= (-I_{N-1}, I_{N-1})^T \otimes \bar{I}_T, \end{aligned}$$

where $\bar{I}_k$ is a $k \times 1$ vector of ones, and $\otimes$ represents the Kronecker product.

To estimate the varying coefficient and trend functions by the LLDVE method, we define the augmented time-varying vector as $\beta_*(\tau) = [f(\tau), \beta_1(\tau), \dots, \beta_d(\tau)]^T$, where β_* is assumed to have first and second order derivatives. After Taylor expansion, we have

$$\beta_*(t/T) \approx \beta_*(\tau) + \beta'_*(\tau)(t/T - \tau),$$

where $\beta'_*(\tau)$ is the first derivative of $\beta_*(\tau)$. It follows that

$$\tilde{f} + \tilde{B}(x, \beta) \approx \tilde{M}(\tau) [\beta_*(\tau) (\beta'_*(\tau))^T]^T,$$

where $\tilde{M}^T(\tau) = [M_1^T(\tau), M_2^T(\tau), \dots, M_N^T(\tau)]^T$ with

$$M_i(\tau) = \begin{pmatrix} 1 & X_{i1}^T & \frac{1-\tau T}{Th} & \frac{1-\tau T}{Th} X_{i1}^T \\ \vdots & \vdots & \vdots & \vdots \\ 1 & X_{iT}^T & \frac{T-\tau T}{Th} & \frac{T-\tau T}{Th} X_{iT}^T \end{pmatrix},$$

for $i = 1, 2, \dots, N$.

Estimation of the parameters is performed by minimizing the weighted least squares given in Eq. (3) with respect to α and $(\beta_*(\tau), h\beta'_*(\tau))^\top$. Thus, we have

$$\min \left[\tilde{Y} - \tilde{M}(\tau)(\beta_*(\tau), h\beta'_*(\tau))^\top - D\alpha \right]^\top \tilde{W}(\tau) \left[\tilde{Y} - \tilde{M}(\tau)(\beta_*(\tau), h\beta'_*(\tau))^\top - D\alpha \right], \quad (3)$$

where $\tilde{W}(\tau) = I_N \otimes W(\tau)$ is the kernel weight matrix with

$$W(\tau) = \text{diag} \left\{ \frac{1}{h} K \left(\frac{1 - \tau T}{Th} \right), \dots, \frac{1}{h} K \left(\frac{1 - \tau T}{Th} \right) \right\},$$

$K(\cdot)$ is a kernel function and h is the bandwidth.

Taking the first order condition of Eq. (3) with respect to α , we obtain, $\hat{\alpha}_\tau = [\tilde{D}^\top \tilde{W}(\tau) \tilde{D}]^{-1} \tilde{D}^\top \tilde{W}(\tau) [\tilde{Y} - \tilde{M}(\tau)(\beta_*(\tau), h\beta'_*(\tau))^\top]^\top$.

Replacing α with $\hat{\alpha}$ in Eq. (3) and minimizing with respect to $(\beta_*(\tau), h[\beta'_*(\tau)]^\top)^\top$ yields the LLDVE estimates, $\hat{\beta}_*(\tau)$ of $\beta_*(\tau)$, given by

$$\hat{\beta}_*(\tau) = (I_{d+1}, \mathbf{0}_{d+1}) \left(\tilde{M}^\top(\tau) \tilde{W}^*(\tau) \tilde{M}(\tau) \right)^{-1} \tilde{M}^\top(\tau) \tilde{W}^*(\tau) \tilde{Y}, \quad (4)$$

where I_{d+1} is a $(d+1) \times (d+1)$ identity matrix, $\mathbf{0}_{d+1}$ is a $(d+1) \times (d+1)$ matrix of zeros, $\hat{\beta}_*(\tau) = [\hat{f}(\tau), \hat{\beta}_1(\tau), \dots, \hat{\beta}_d(\tau)]^\top$, $\tilde{W}^*(\tau) = \tilde{K}^\top(\tau) \tilde{W}(\tau) \tilde{K}(\tau)$, with $\tilde{K}(\tau) = I_{NT} - D(D^\top \tilde{W}(\tau) D)^{-1} D^\top \tilde{W}(\tau)$. Note that $\tilde{K}(\tau) D\alpha = 0$ for any τ .

We estimate individual trend functions from the vector of residuals

$$\hat{e}_{it} = Y_{it} - \hat{f}(t/T) - X_{it}^\top \hat{\beta}_t - \hat{\alpha}_i.$$

Specifically, for the i^{th} cross-section, the trend function, $\hat{m}_i(t/T)$, is estimated by a local linear regression of $\hat{e}_{it}$ on $\tau = t/T$:

$$\hat{e}_{it} = m_i(t/T) + \varepsilon_{it}, \quad (5)$$

where ε_{it} , for $t = 1, 2, \dots, T$, are independent errors with a zero mean.

Following Sun et al. (2009) and Silvapulle et al. (2017), we adopt the leave-one-unit-out cross-validation method of bandwidth selection. This procedure automatically chooses the optimal bandwidth. We use a bootstrapping method to construct nonparametric confidence bands for the time-varying trend and coefficient functions following Mammen (1993) and Silvapulle et al. (2017). The first step is to obtain the detrended residuals from Eq. (5), $\hat{e}_{it} = \hat{e}_{it} - \hat{m}_i(\tau; b)$, where $\hat{e}_i = (\hat{e}_{i1}, \dots, \hat{e}_{iT})$ for $i = 1, 2, \dots, N$. Next, we re-sample the detrended residuals $\hat{e}_{it}^* = \hat{e}_{ik}$, where k is randomly selected from $\{1, \dots, T\}$ with replacement and generate a bootstrapping sample of Y_{it}^* through

$$Y_{it}^* = \hat{f}(1/T) + X_{it}^\top \hat{\beta}_t + \hat{\alpha}_i + \hat{m}_i(\tau; b) + \hat{e}_{it}^*, \quad \text{for } t = 1, 2, \dots, T.$$

Using the bootstrapped sample of $\{Y_{it}^*, X_{it}\}$, we get the estimates of time-varying common trend functions $\hat{f}(t/T)$, the coefficient functions $\hat{\beta}_t^*$ and the individual trend functions by the LLDVE method. The final step is to repeat the above procedure 1000 times and obtain the 90% confidence bands for the estimates of the common trend and coefficient functions.

2.2. Endogeneity

Oil prices and economic policy uncertainty might be simultaneously determined, which may pose a challenge to pin down the

direction of causality. To address this issue, we employ a procedure for consistent estimation of nonparametric simultaneous equations models proposed by Su and Ullah (2008). The basic idea of the proposed nonparametric estimator is similar to standard two-stage least-squares. In the first stage, we run a nonparametric regression of the endogenous variable on all the exogenous regressors. In the second stage, we run a nonparametric regression of the dependent variable on each of the regressors, including the endogenous variable and the residuals from the first stage regression. The third stage involves marginal integration to ensure that the assumption of a zero mean error term holds. To briefly explain the Su and Ullah (2008) estimator, consider a model that treats the real price of oil as the endogenous regressor:

$$Y_i = m(X_i, Z_{1i}) + \varepsilon_i, \quad \text{and} \quad X_i = g(Z_i) + u_i,$$

where Y_i is economic policy uncertainty, $m(\cdot)$ is the unknown smooth function of interest, X_i is the endogenous regressor (real price of oil), $Z_i = (Z_{1i}, Z_{2i})$ where Z_{1i} and Z_{2i} are $d_1 \times 1$ and $d_2 \times 1$ vectors of instrumental variables respectively, $g(\cdot)$ is an unknown smooth function of the instruments Z and u are disturbances. We assume that $E(u|Z) = 0$ and $E(\varepsilon|Z, u) = E(\varepsilon|u)$. The model allows for both u and ε to be heteroscedastic. Su and Ullah (2008) show that the local linear least square (LLLS) estimation of $m(\cdot)$ is feasible and can be estimated as follows.

$$\begin{aligned} E(Y|X, Z, u) &= m(X, Z_1) + E(\varepsilon|X, Z, u) = m(X, Z_1) + E(\varepsilon|X - g(Z), Z, u), \\ &= m(X, Z_1) + E(\varepsilon|Z, u) = m(X, Z_1) + E(\varepsilon|u). \end{aligned}$$

By the law of iterated expectations, and since $Z_1 \in Z$, we have $w(X, Z_1, u) \equiv E(Y|X, Z_1, u) = E(Y|X, Z_1, u) = m(X, Z_1) + E(\varepsilon|u)$. Since u are unobserved, Su and Ullah (2008) propose that they can be replaced by the residuals obtained by regressing X on Z nonparametrically to provide consistent estimates of $m(X, Z_1)$ without further restrictions.

Assuming that $E(\varepsilon) = 0$, the procedure for the Su and Ullah (2008) estimator is as follows:

Step 1. Run a local-constant regression of X on Z with kernel function $K_{1,h}(\cdot)$ and bandwidth vector h_1 to get a consistent estimate, $\hat{g}(Z)$ of $g(\cdot)$ and obtain the residuals $\hat{u}_i = X_i - \hat{g}(Z_i)$, for $i = 1, 2, \dots, n$.

Step 2. Run a local linear regression of Y on X, Z_1 and $\hat{u}$ with kernel function $K_{2,h}(\cdot)$ and bandwidth vector h_2 to get a consistent estimate, $\hat{w}(X, Z_1, u)$ of $w(\cdot)$.

Step 3: Obtain a consistent estimate of $m(\cdot)$ as

$$\hat{m}(X, Z_1) = \frac{1}{n} \sum_{i=1}^n \hat{w}(X, Z_1, \hat{u}_i)$$

where $\hat{w}(X, Z_1, \hat{u}_i)$ is the counterfactual estimate of $\hat{w}(X, Z_1, u)$ obtained using the bandwidths from the local linear regression in step 2 and assuming that $E(\varepsilon) = 0$. The derivatives of $m(\cdot)$ are given by

$$\hat{m}'(X, Z_1) = \frac{1}{n} \sum_{i=1}^n \hat{w}'(X, Z_1, \hat{u}_i)$$

where $\hat{w}'(X, Z_1, \hat{u}_i)$ is the counterfactual derivative of $w(\cdot)$.

As outlined in the application of the method by Henderson et al. (2013), the instrumental variable estimator of Su and Ullah (2008) is an appropriate applied nonparametric econometric tool to deal with potential endogeneity in kernel based nonparametric methods.

2.3. Parametric panel data model

As a benchmark exercise, we use the Common Correlated Effects Mean Group (CCEMG) estimator proposed by Chudik and Pesaran (2015) to motivate the rationale for our use of the nonparametric method. The dynamic CCEMG estimator is appealing as it accounts for key issues in panel data; namely, endogeneity, heterogeneity and cross-sectional dependence.

Following Chudik and Pesaran (2015), the dynamic heterogeneous panel data model with a multifactor error term that accommodates an unobserved factor structure is given by:

$$Y_{it} = \alpha_i + \phi_i Y_{i,t-1} + \beta_i X_{it} + u_{it}, \quad u_{it} = \gamma' f_t + e_{it}, \quad (6)$$

where f_t denotes a vector of m -dimensional unobserved common factors, γ_i is an $m \times 1$ vector heterogeneous factor loading and e_{it} is a country-specific error term. Chudik and Pesaran (2015) propose that Eq. (6) can be consistently estimated using the dynamic CCEMG estimator by including sufficient number of lags of cross-section averages as follows.

$$Y_{it} = \alpha_i + \phi_i Y_{i,t-1} + \beta_i X_{it} + \sum_{l=0}^p \delta_{il} \bar{Z}_{t-l} + \epsilon_{it}, \quad (7)$$

where $\bar{Z}_t = (\bar{Y}_t, \bar{X}_t)$, with $\bar{Y}_t$ and $\bar{X}_t$ denoting simple cross-sectional averages of Y_{it} and X_{it} in year t . The mean group estimates are given by $\hat{\gamma}_{MG} = \frac{1}{N} \sum_{i=1}^N \hat{\gamma}_i$ with $\hat{\gamma}_i = (\hat{\phi}_i, \hat{\beta}_i)$. This estimator allows the error term to be correlated serially, as well as cross-sectionally, and it is robust to endogeneity. The error correction model (ECM) corresponding to Eq. (8) is given by:

$$\Delta Y_{it} = \lambda(Y_{i,t-1} - \theta_i X_{it}) + \delta_{0,i} + \delta_{1,i} \Delta X_{it} + \epsilon_{it}, \quad (8)$$

where δ_i are heterogeneous short-run coefficients and θ_i are the long-run coefficients assumed to be homogeneous across countries. The coefficient of ECM, λ_i , measures the speed of adjustment to long-run equilibrium after a shock. As outlined in Pesaran et al. (1999), the mean group estimator based on the autoregressive distributed lag approach is suitable for long-run analysis and it is robust to endogeneity of regressors. In addition, they show that the estimator is valid irrespective of whether the series are $I(0)$ or $I(1)$.

3. Data and benchmark results

3.1. Data

To measure economic policy uncertainty, we use the monthly indices constructed by Baker and colleagues as described in Baker et al. (2016), but since updated beyond 2016. Baker and colleagues constructed policy-related economic uncertainty indices for a number of countries based on newspaper coverage frequency. The indices capture uncertainty related to the decision makers, economic policy actions and the timing and effect of economic policies. To construct the indices in each country, they consider major newspapers for each country and search the digital archives of each newspaper to obtain a monthly count of articles that contain terms describing the economy, policy and uncertainty. To account for variations in the overall volume of articles across newspapers and time, they scale the raw counts by the total number of articles in the same newspaper/month and standardize each monthly paper series to unit standard deviation over the period under consideration. They then compute the average across the newspapers considered by month and normalize the

series by the mean (for full details on construction of the indices see Baker et al., 2016).

Our oil price variable is West Texas Intermediate (WTI) crude oil prices (Dollars per Barrel), obtained from the US Energy Information Administration (EIA). Following the common practice of the EIA, we transformed the nominal WTI oil prices to real terms by dividing the monthly nominal prices by the ratio of the Consumer Price Index (CPI) in that month to the CPI in a base month. We use daily data on WTI crude oil prices to compute oil price volatility. We computed oil price volatility for each month as the sample standard deviation of percentage changes in daily crude oil prices over the trading days, $\sqrt{\frac{\sum_{i=1}^N (p_i - \bar{p})^2}{N-1}}$, where p_i and $\bar{p}$ are the daily percentage changes in WTI oil price and the average change during the month, respectively and N denotes the number of days in each month. Data on indices of industrial production and long-term real interest rates are sourced from the OECD database.

Table 1 shows the summary statistics for the series. As shown in Table 1, economic policy uncertainty indices vary significantly in the G7 countries, with overall standard deviation of 84.56. The maximum value of economic policy uncertainty index in the sample is recorded for the UK, reflecting the uncertainty generated by the Brexit referendum. As shown in Table 1, the real price of oil also varies significantly with a standard deviation of \$30.7 per Barrel.

Fig. 1 shows that the economic policy uncertainty index increased sharply with the onset of the GFC and intensified with the increase in oil price volatility and the euro area debt crisis in 2011.

Economic policy uncertainty remained elevated even after the GFC, except for the rollback in the trend between 2012 and 2014. The uncertainty index hit a new all-time high in 2016 with the Brexit referendum where the index was almost five times the value just before the GFC. The economic policy uncertainty index also increased sharply in the US following the last presidential election due to uncertainties surrounding various policies to be undertaken by the new administration. The indices for Europe and the US exhibit similar trends with some common peaks, such as during the 2008/2009 GFC and during the European debt crisis in the second half of 2011. Oil prices also significantly increased during these periods before they plunged from 2014 onwards.

3.2. Benchmark parametric estimation results

We pre-test for the presence of cross-sectional dependence (CSD) in our series using the CSD test proposed by Pesaran (2004). As shown in Table 2, the null hypothesis of cross-sectional independence is rejected at the 1% level of significance, suggesting the need to employ the dynamic CCEMG estimator that accounts for cross-sectional dependence in errors.

Table 3 presents the baseline parametric results from the dynamic CCEMG estimator. Column 1 presents the estimates using the full sample while column 2 and column 3 present the estimates using the subsamples for the pre- and post-GFC periods, respectively. Our results provide some evidence of a time-varying relationship between economic policy uncertainty and oil prices. The coefficient on real oil price is negative and statistically significant in the pre-GFC

Table 1
Summary statistics: monthly data (1997:01–2018:06; and observations = 1806).

Variable	Mean	Standard deviation	Minimum	Maximum
Economic policy uncertainty (index)	130.24	84.56	11.29	1141.80
Real price of oil (WTI – Dollars per Barrel)	66.48	30.71	17.50	156.20
Oil price volatility	2.26	1.04	0.71	7.64
Index of industrial production	102.43	8.88	76.68	122.77
Real interest rate (%)	3.33	1.72	–0.23	7.87

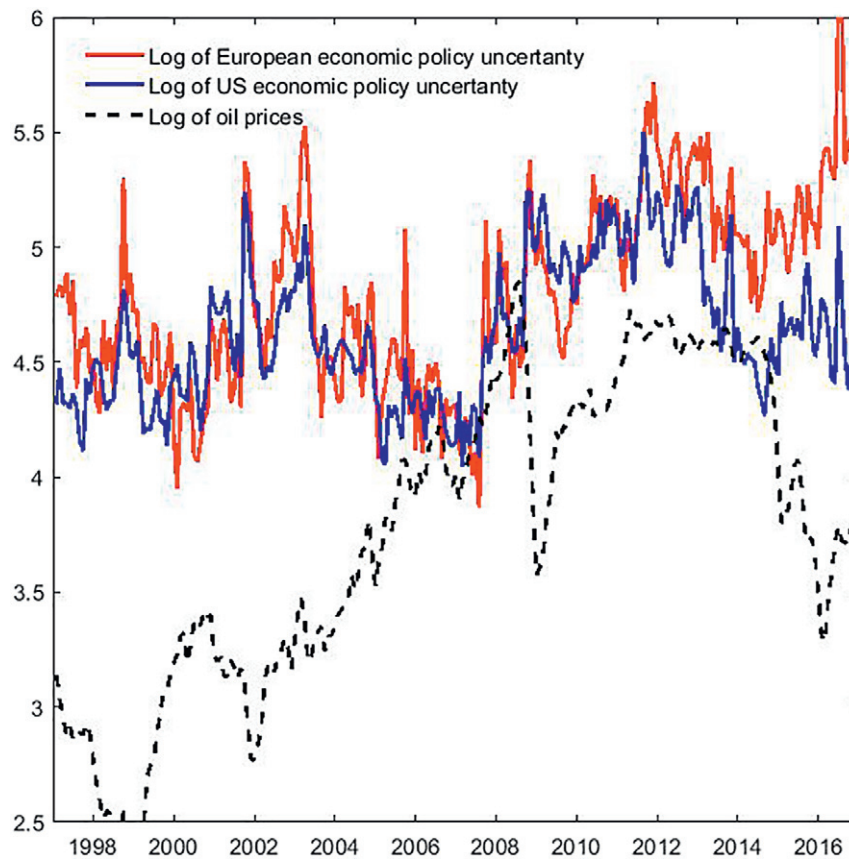


Fig. 1. Trends in oil price volatility and indices of economic policy uncertainty.

period. The effect of oil price on economic policy uncertainty in the post-GFC period is positive, but is statistically insignificant at conventional significance levels. More precisely, our estimates show that the point elasticity of economic policy uncertainty with respect to changes in oil prices is about -0.43 in the pre-GFC period. The interpretation of the coefficient -0.43 is that, on average, a 1% increase in real oil prices is associated with a 0.43% decrease in economic policy uncertainty during the pre-GFC period.

The difference in the impact of oil prices on economic policy uncertainty in the pre-GFC and post-GFC periods can be attributed to the sources of oil price increases in the respective periods. In particular, the negative effect of oil prices in the pre-GFC period can be explained by the surge in global aggregate demand that occurred at this time that drives oil prices and tends to reduce uncertainty. Our results are in line with the findings in existing literature that documents the positive effect of aggregate demand on oil prices before the GFC, which created a conducive environment for the global economy, instead of triggering recessions (see e.g. Kilian, 2009). Along these lines, our results indicate that the rise in oil prices during periods of booms in aggregate demand may dampen economic policy uncertainty, thereby creating a positive outlook for the economy.

Table 2
Pesaran (2004) test for cross-sectional dependence in the ADF(p) regression.

Variable	CD-test statistic	p -Value
Economic policy uncertainty	38.36	0.00
Real price of oil	73.61	0.00
Oil price volatility	73.61	0.00
Index of industrial production	22.11	0.00
Real interest rate	64.10	0.00

Note: Under the null hypothesis of no cross-sectional dependence, $CD \sim N(0, 1)$. The p -values are zero indicating dependence across countries.

To capture the difference between pre- and post-GFC periods, we include a dummy variable for the global financial crises that takes the value of 1 for the GFC period between July 2007 and December 2008 and 0 otherwise. The coefficient of the GFC dummy variable is positive and statistically significant, suggesting that economic policy uncertainty increases in periods of economic crises. Specifically, Table 3 shows that the index of economic policy uncertainty is higher by 0.21 log points (1.23) during the period of economic crises compared to the period without crises.

The condition for the existence of a dynamically stable long-run relationship between economic policy uncertainty and oil prices is that the coefficient on the ECM term be negative and within the unit circle. As shown in Table 3, the coefficients on the ECM

Table 3
Baseline parametric estimates.

	(1) Full sample	(2) Pre-GFC	(3) Post-GFC
Real price of oil	0.075 (0.110)	-0.431^{***} (0.128)	0.015 (0.106)
Oil price volatility	0.0990^{***} (0.034)	0.201^{***} (0.070)	-0.078 (0.073)
Index of industrial production	-1.973^{**} (1.000)	-7.300^{**} (3.001)	2.148 (1.674)
Real interest rate	-0.234^{***} (0.065)	0.014 (0.089)	-0.266^{***} (0.083)
Dummy for GFC	0.210^{***} (0.071)		
ECM	-0.575^{***} (0.011)	-0.418^{***} (0.039)	-0.419^{***} (0.024)
R ²	0.324	0.259	0.235
Observations	1792	868	791

Note: The dependent variable is the log of economic policy uncertainty. Standard errors in parenthesis. *** and ** denote significance at 1% and 5% levels, respectively.

falls within the dynamically stable range. The estimate of the ECM coefficient captures the speed of adjustment to long-run equilibrium. For example, the interpretation of the ECM coefficient of -0.575 in column (1) is that the system is converging to equilibrium by 57.5% and that the estimated model is stable. The R^2 shows that a sizable proportion of the variations in economic policy uncertainty is explained by our model. For example, the estimate in column (1) shows that about 32.4% of the variation in economic policy uncertainty is explained by our model.

A potential shortcoming of the parametric panel data estimator is that the estimates may be inconsistent if the model is misspecified. To address this issue, and to capture the smooth time-varying relationship between oil prices and economic policy uncertainty, we employ local linear estimators of a varying coefficient nonparametric panel data technique.

4. Nonparametric estimation of varying coefficients and trend functions

Before we estimate the trend and coefficient functions, we test for the existence of country-specific and global trends in Section 4.1.

4.1. Nonparametric trend tests

We adopted the Mann-Kendall (MK) nonparametric trend test (Mann, 1945; Kendall, 1975) which is the most common test employed to detect monotonic trends in energy and climate data series. An advantage of the nonparametric MK test is that it does not require the data to be normally distributed. In addition, the MK test is robust to abrupt breaks in data series and invariant to transformations, such as logs. Thus, it is a superior test to detect the existence of linear or nonlinear trends. The null hypothesis is that the data series are independent and identically distributed, whereas the alternative hypothesis is that the data series follow a monotonic trend. If X_i and X_j are the sequential data values of the time series at time i and j , where $j > i$ we compute the difference, $X_j - X_i$ and assign value of 1, 0, or -1 to positive, zero and negative differences, respectively. The MK test statistic is then computed as the sum of the integers:

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{sgn}(X_j - X_i),$$

where $\text{sgn}(X_j - X_i)$ equals 1, 0, or -1 as defined above.

That is, the MK test analyzes the sign of the difference between the later-measured value X_j and all earlier-measured values, X_i . Large positive values of S indicate an upward trend, while large negative values of S suggest a downward trend. The statistic S has an approximate normal distribution with the following Z-transformation.

$$Z = \begin{cases} \frac{S-1}{\sigma}, & \text{if } S > 0 \\ 0 & \text{if } S = 0, \\ \frac{S+1}{\sigma} & \text{if } S < 0 \end{cases}$$

where $\sigma = \left(\frac{1}{18} \left[n(n-1)(2n+5) - \sum_{j=1}^p (t_j-1)(2t_j+5) \right] \right)^{1/2}$ is the standard deviation of S , n is the length of the time series, p is the number of tied groups (observations of equal values) in the data series and t_j is the number of data points in the j^{th} tied group. If $Z > 0$, trends are increasing, while if $Z < 0$ trends are decreasing. The null hypothesis of no trends is rejected if $|Z| > Z_{1-\alpha/2}$, suggesting that a significant trend exists in the time series.

Table 4 presents the nonparametric MK test results. The Table shows that economic policy uncertainty (EPU) exhibits a positive and significant trend in all countries except for Japan. The global trend in economic policy uncertainty is also positive and significantly different from zero.

Oil price is a global variable common to all countries. The nonparametric trend test indicates that the crude oil price exhibits a significant positive trend. The trend in the index of industrial production is positive in some countries and negative in others with a positive overall global trend. The real interest rate shows a significant negative trend in all G7 countries. Having confirmed the existence of trends in our series, we proceed to a more rigorous nonparametric analysis of the time varying coefficient and trend functions.

4.2. Local linear estimates

In this section we present the estimates of $f(\cdot)$ and $\beta_j(\cdot)$ for the observed data, $\{(Y_i, X_i)\}$, using the LLDVE method discussed in Section 2. Fig. 2 presents the results from the local linear dummy variable estimates along with 90% confidence intervals. Our nonparametric panel data estimation captures the time-varying relationship between oil prices and economic policy uncertainty. Specifically, oil prices and economic policy uncertainty exhibit a negative relationship before 2010. The coefficient on the oil price increases sharply following the GFC and becomes positive with the onset of the Arab Spring in 2011, before becoming negative and significant from 2016 onwards. The main reason for the negative relationship between oil prices and economic policy uncertainty is that the main driver of oil prices since the late 1990s has been a surge in aggregate demand, rather than supply shocks (Kilian, 2009). In sum, our local linear estimates show that the coefficient function on oil price volatility was largely insignificant most of the period before the GFC and strongly significant and positive following the GFC. Our findings are in line with existing literature. Specifically, our results support the findings of Kang et al. (2017) and Antonakakis et al. (2014) in the sense that spillovers between oil prices and economic policy uncertainty increased significantly during the GFC of 2008–2009. Another striking feature observed in Fig. 2 is that the effect of oil price volatility on economic policy uncertainty increased modestly following Hurricane Katrina in 2005 and then dramatically following the Arab Spring in 2011.

Our results also show that there is a time-varying relationship between oil price volatility and economic policy uncertainty. Specifically, oil price volatility and economic policy uncertainty have a significant positive relationship for the period between 2001 and 2014. Higher oil price volatility is associated with elevated uncertainty as it affects expected profits of firms, thus leading to the

Table 4
Nonparametric country-specific and global trend tests.

	CA	DEU	FRA	GBR	ITA	JPN	USA	Global
EPU	11.61	8.87	14.26	14.42	2.04	−0.70	5.96	10.22
Industrial production	7.61	18.80	−3.76	−11.16	−10.50	−1.50	14.78	3.23
Real interest rate	−19.54	−18.84	−18.38	−18.36	−11.51	−15.87	−16.87	−18.33
Real price of oil								10.82
Oil price volatility								−4.82

Note: The critical value, $Z_{1-\alpha/2}$, for the 5% significance level is 1.96. Thus, if $|Z| > 1.96$, it suggests rejection of the null hypothesis of no trend in the data series. Country codes: CA = Canada, DEU = Germany, FRA = France, GBR = United Kingdom, ITA = Italy, JPN = Japan, USA = United States.

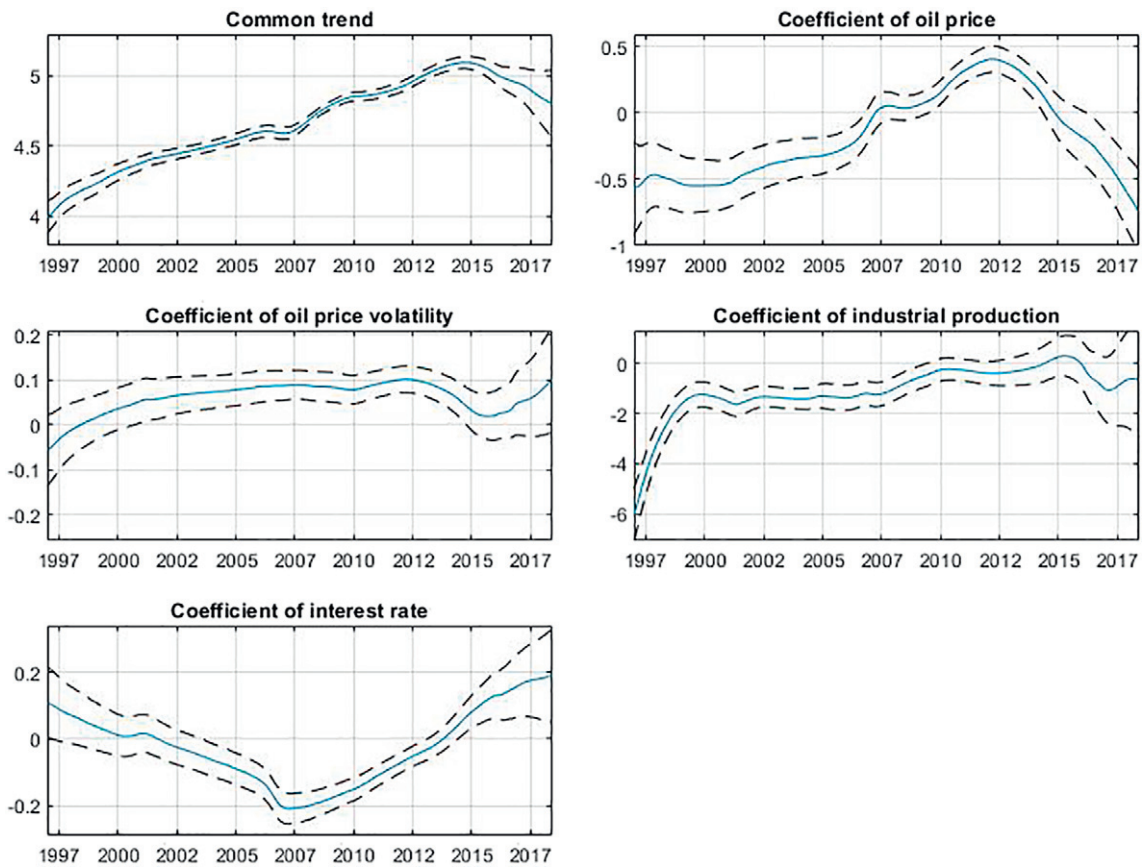


Fig. 2. Estimates of common trend and coefficient functions with 90% confidence bands.

decision by firms to postpone investments, which creates mounting pressure on policymakers to respond with appropriate policies. The coefficients on industrial production and real interest rates have the expected signs, which are negative and statistically significant for much of the sample period. The negative correlation between the real interest rate and the index of economic policy uncertainty is consistent with the idea that before, or during, recessions investors shift away from risky assets to safe assets, in order to avoid short-term losses. This causes an increase in the price of safe assets, such as Treasury Bills. The rise in the price of safe assets translates into a decline in nominal interest rates. With inflation being sluggish in the short-run, this causes a decline in the real interest rate. Investor flight to safe assets, and the associated actions from policymakers, generate an increase in economic policy uncertainty.

The effects of changes in oil prices might be different for net oil exporters and importers. Higher oil prices are expected to have a positive effect for net oil exporting economies, resulting in lower uncertainty. This is because of the potential improvement in terms of trade from boosting oil revenue. In addition, if the increase in oil price is being driven by a demand shock, we would expect a decrease in economic policy uncertainty in oil importers and exporters, but if it is a supply shock, an oil price rise might increase economic policy uncertainty in oil importers. Since Canada is the only net oil exporter among the G7 countries, we conduct a sensitivity check by re-estimating our model excluding Canada from our sample. Fig. 3 presents the estimated coefficient of oil price volatility and the common trend functions. Our results remain robust to the exclusion of Canada. This is a plausible result because the increase in oil prices since the late 1990s has been mainly driven by a positive aggregate demand shock.

What drives the time-varying relationship between oil price volatility and economic policy uncertainty? Several factors can be behind the underlying time-varying relationship between the two variables over time. First, the increase in the price of oil can be driven by either a disruption in crude oil supply or by the surge in aggregate demand driven by a booming world economy. The main reason that oil price volatility did not cause an increase in economic policy uncertainty in the pre-GFC period was that the surge in oil prices in this period was driven by an increase in aggregate demand for global commodity prices. As Kilian (2009) shows, the increase in oil prices after 1979, and particularly after 2003, has not resulted in recessions since oil price rises in this period were driven primarily by strong global aggregate demand stemming from a booming world economy. In particular, the high demand for crude oil from emerging-market economies in recent decades has been the primary source of increases in oil prices. Notable studies on the relationship between oil prices and economic policy uncertainty provide evidence in line with this analysis (see e.g., Antonakakis et al., 2014; Kang and Ratti, 2013a,b; Kang et al., 2017). In a study of the causal relationship between oil prices and economic policy uncertainty, Kang and Ratti (2013a,b) find that positive aggregate demand shocks lead to a significant negative effect on policy uncertainty. In line with findings in Antonakakis et al. (2014), during the 2008–2009 GFC and 2011 Arab Spring, the coefficient on the real price of oil increased sharply, suggesting its strong influence on fueling economic policy uncertainty.

Studies subsequent to Kilian (2009) by several authors, who have used structural VAR models (see e.g., Peersman and Van Robays, 2012; Lippi and Nobili, 2012), show that an increase in oil prices has an adverse effect on the economy when induced by supply shocks, rather than aggregate demand shocks.

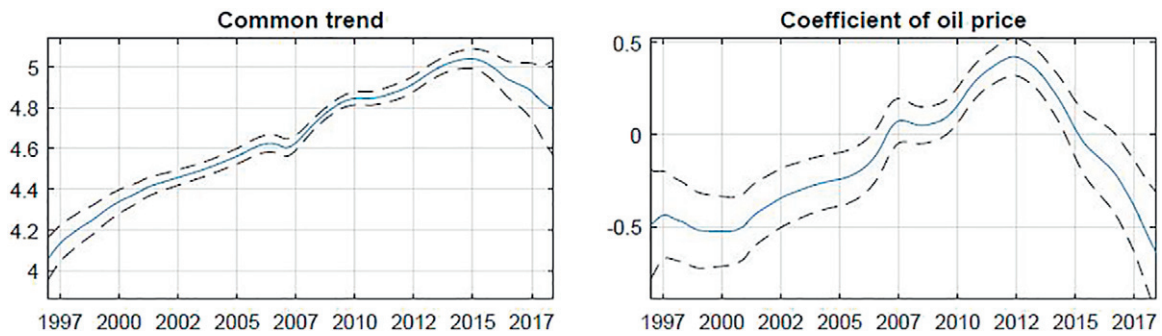


Fig. 3. Estimates of coefficient of oil price volatility and the common trend function for G7 countries excluding Canada.

Studies that examine the direct relationship between oil prices and economic policy uncertainty (e.g., Antonakakis et al., 2014; Kang and Ratti, 2013a,b) also show that the effects of an increase in oil prices that arise from an unanticipated increase in global aggregate demand shocks are very different from those caused by a disruption in oil supply or an unanticipated increase in oil prices due to precautionary demand, resulting from fears about future shortages in crude oil supply. As noted in Kilian (2008), economic growth in Japan was above the normal rate following the surge in oil prices between 2002 and the GFC, due to the expansion of the global economy driven by an increase in aggregate demand that counteracted the negative effects of higher oil prices and reduced economic policy uncertainty. Our findings with respect to the time-varying relationship between oil price volatility and economic policy uncertainty is consistent with the argument in Kilian (2009) that the effect of an increase in oil price in a particular period depends on the underlying reason for the oil price increase. Our findings are consistent with recent studies that have applied time-varying parameter VARs and concluded that prevailing economic conditions in each period matter for the oil price-economic policy uncertainty relationship (Degiannakis et al., 2018). Specifically, the surge in global demand for industrial commodities from the early 2000s until the GFC, stemming from a boom in global economic activity, dampened the effect of the oil price increase on economic policy uncertainty during this period.

Another explanation for the negative relationship between oil prices and economic policy uncertainty could be changes in the transmission mechanism of shocks. Specifically, some of the negative effects of higher oil prices on the economy are mitigated by the substitution to high quality fuels and continued technological advancement that have given rise to energy efficiency gains over past decades. Blanchard and Gali (2007) and Blanchard and Riggi (2013) suggest that the mechanism by which oil price shocks transmit to macroeconomic effects can change with the structure of the economy and policy framework. Substantial oil price declines since the summer of 2014 were driven by large-scale investment and technological innovation that gave rise to the shale oil and gas revolution in the US. The fracking boom in the US has increased the supply of oil, which has contributed to the lower oil price (Bataa and Park, 2017; Kilian, 2017). Substitution to natural gas, associated with the shale gas revolution has also contributed to the decline in oil prices since 2014 (Middleton et al., 2017). The natural gas market is the biggest competitor of crude oil, especially in the transportation sector. This reflects global concerns over the long-term impact of carbon emissions from crude oil and the fact that natural gas is the cleanest energy source among fossil fuels.

4.3. Common and country-specific trend functions

We use our nonparametric panel data model to decompose the time-varying trends of economic policy uncertainty into

country-specific and common trends. Our nonparametric varying-coefficient panel data model with a local linear dummy variable estimation (LLDVE) technique enables us to estimate the trend functions in a highly nonlinear fashion. Unlike the techniques employed in the existing literature, our method allows the common trend functions to evolve over time to capture common global shocks. More importantly, the nonparametric panel data model also allows for time-varying coefficient functions, as well as country-specific and common trend functions to capture nonlinearity and heterogeneity across time and countries. Our estimates of these trends functions shed light on long-term movements in economic policy uncertainty and their implications for long-term policy directions.

The LLDVE estimates of country-specific trend functions, along with the common trend function, are plotted in Fig. 4. We present the country-specific economic policy uncertainty trend functions and the common trend functions together for comparison. The dotted lines represent the 90% confidence bands around the common trend functions. As shown in Fig. 4, the common trend has increased sharply since the beginning of the sample period and peaked in 2015. The trend has decreased modestly thereafter. We find that the country-specific trends largely follow the common component suggesting that global factors play an important role in driving the time-varying increasing trend in country-specific uncertainty.

Different patterns are observed for the country-specific economic policy uncertainty trend functions. Fig. 4 shows that the trend function for the economic policy uncertainty index in Canada has increased since 2006 and has been above the global trend since 2010. The trend function for France has been above the global trend since 2003 and continued to be upward trending except at the very end of the study period. For Italy and Germany, the country-specific trend functions have been below the global trend since 2005, except for the period during the European debt crisis in 2011 to 2012. For Japan, the country-specific trend function jumps above the global trend at the onset of the GFC, but this is reversed after 2010, while in the US, the country-specific trend function has been largely below the global trend except for the period before 2005 and during the GFC. Overall, the country-specific trend function for economic policy uncertainty has been largely following the global trend function for the G7 countries.

Our results support findings of earlier studies. Specifically, our findings lend support to Berger et al. (2016) and Mumtaz and Theodoridis (2017) that have documented co-movements, common and country-specific trends in uncertainty, in which the common component has become more important in recent decades. These earlier studies provide evidence of the role that common movements in uncertainty play in driving country-specific trends. Our findings also suggest important policy implications, highlighting that policy makers need to consider the role of globalization by accounting for economic and political factors, such as the Euro Area debt crisis that may cause country-specific uncertainty.

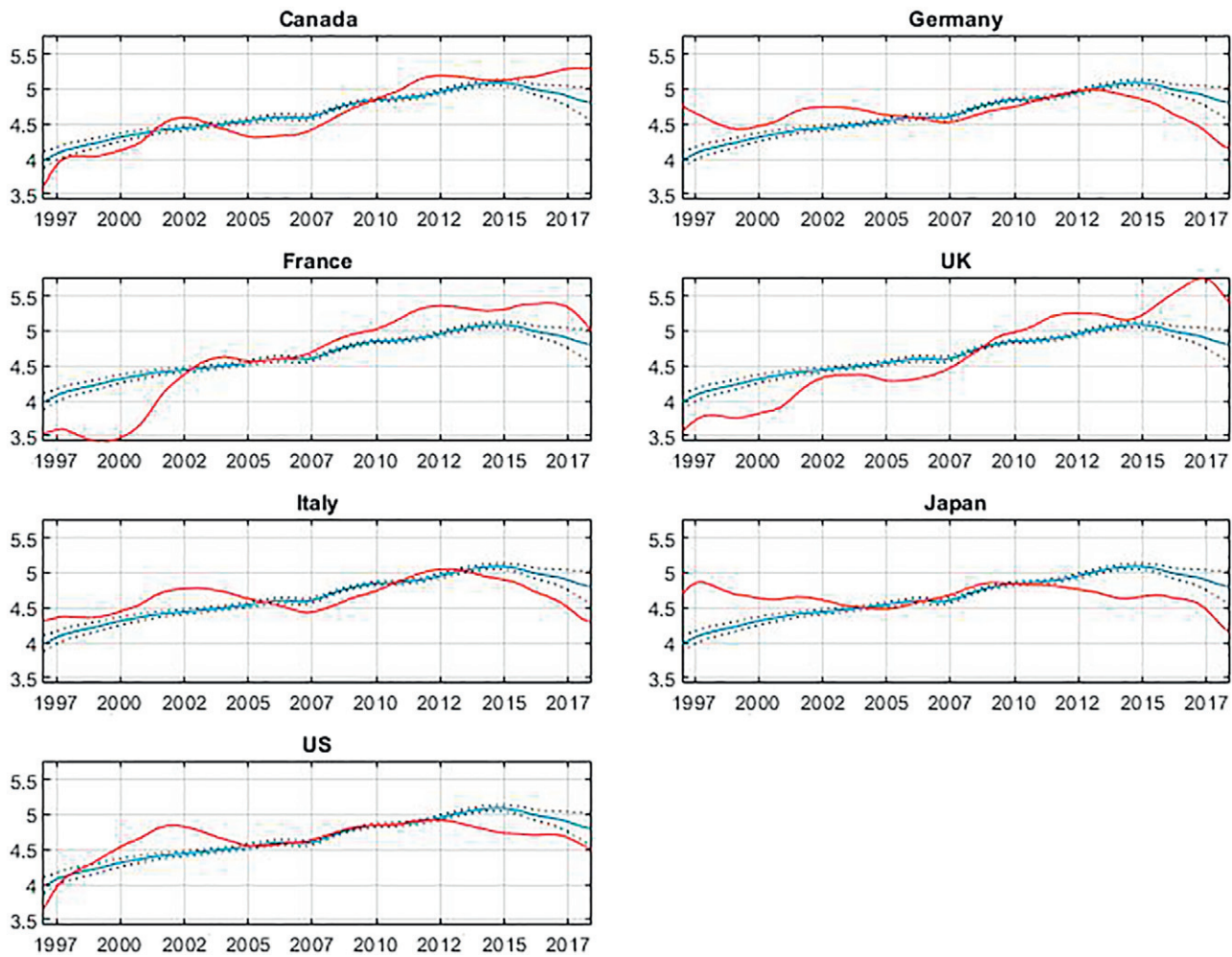


Fig. 4. Estimates of country-specific trend functions (solid red) and the common trend function (solid blue) with 90% confidence bands (dotted black).

4.4. Robustness checks

4.4.1. Endogeneity and nonlinearity in the nonparametric estimates

We conduct alternative estimations to check for endogeneity and nonlinearity, employing the Su and Ullah (2008) estimator outlined in Section 2.2. We model economic policy uncertainty allowing for the real oil price to be the endogenous regressor instrumented by its lagged value. Following Henderson and Parmeter (2015), we compare the 45° gradient plots (Fig. 5) for the local linear estimates that ignore endogeneity of real oil prices and the instrumental variable local linear estimates to visualize the effect of accounting for endogeneity and to detect nonlinearity in our estimates. The basic idea of the 45° plots is as follows. Each point on the dark black line represents the estimated gradient, one for each observation. Each point on the red and blue lines in Fig. 5 represents the upper and lower bounds of a 95% bootstrapped confidence interval for the specific point estimates, respectively. The estimated gradient for an observation is positive and statistically significant if the upper and lower confidence bands for that estimate are both in the upper right quadrant. If the confidence bands for a given estimate are both in the lower left quadrant, then the estimate for that observation is negative and statistically significant. The gradient estimate is statistically insignificant if the upper or lower confidence bands straddle the horizontal axis (see Henderson et al., 2012).

As shown in Fig. 5, there is no significant difference between the local linear estimates that ignore endogeneity (panel (a)) and

the instrumental variable local linear estimates that account for endogeneity (panel (b)). The estimated gradients are statistically significant at 5%. They are evenly distributed in the negative and positive quadrants, indicating that oil prices have negative and positive effects on economic policy uncertainty. That is, the relationship between real oil prices and economic policy uncertainty is nonlinear. Our results suggest that failure to account for nonlinearity may lead to inconsistent estimates.

In sum, our results are robust to alternative specifications and endogeneity. More importantly, our results show that it is crucial to account for nonlinearity in modelling the relationship between oil prices and economic policy uncertainty. Although our nonparametric test of endogeneity shows that our estimates are robust to potential endogeneity bias, the fact that the equations are not in a VAR system is still a caveat. As a further robustness check, we estimate the effect of oil price shocks on economic policy uncertainty using a parametric panel VAR model in the next section.

4.4.2. Oil price shock – panel VAR

In this section, we employ a panel VAR (PVAR) model to further examine the dynamic impact and causality between oil price shocks and economic policy uncertainty. The PVAR representation can be written as, $Y_{it} = A_0 + A_1 Y_{it-1} + \theta_i + \gamma_t + e_{it}$, where Y_{it} is a vector variables, $i = 1, \dots, N$, indicate countries, $t = 1, \dots, T$, indicate time (months). θ_i is a vector of country-specific fixed effects,

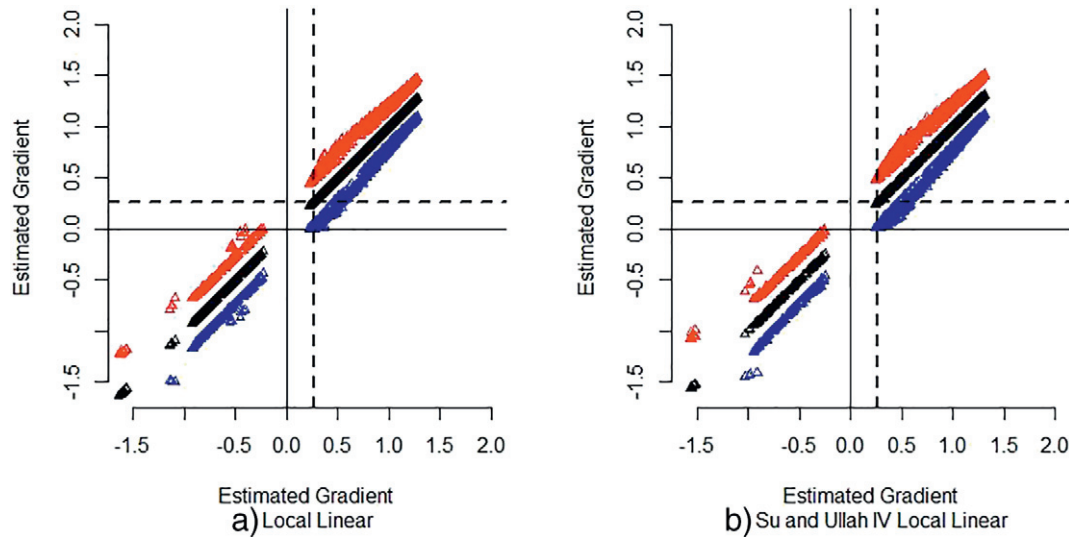


Fig. 5. 45° plot of the estimated gradients for real price of oil.

which controls for the time-invariant unobserved heterogeneity across countries such as geography, history and culture. γ_t denotes a vector of year fixed effects that controls for the shocks common to all countries, such as business cycles. e_{it} denote a vector of disturbance terms that are independently and identically distributed. A_0 and A_1 are coefficient matrices to be estimated. To estimate the PVAR model, we use orthogonality conditions in a generalized method of moments (GMM) framework with a maximum of four lags. We choose optimal lag orders using consistent moment and model selection criteria (MMSC) for GMM models proposed by Andrews and Lu (2001). Because we find strong evidence for nonlinearities in our nonparametric panel data model, we estimate the PVAR model for three regimes: the full sample, pre-GFC and post-GFC regimes.

Based on the three model selection criteria proposed by Andrews and Lu (2001), the smallest values for MMSC-BIC, MMMSC-AIC and MMSC-QIC are obtained when the lag length is 2. Accordingly, we fit a second-order PVAR model with the same specification of instruments as above using GMM estimation. Panel (a), Panel (b) and Panel (c) of Fig. 6 show the response of economic policy uncertainty to each of the shocks using the full sample, pre-GFC sample and post-GFC sample, respectively. The results in Fig. 6 are suggestive of a time-varying relationship between oil prices and economic policy uncertainty, consistent with the nonparametric estimates discussed in Section 4.2. As shown in panel (b) of Fig. 6, oil price shocks have an immediate negative effect in the short-term (up to 3 months) in the pre-GFC period. Panel (c) of Fig. 6 shows that oil price shocks cause an increase in economic policy uncertainty from 4 to 10 months following the shock, whereas oil price volatility has an immediate positive and significant effect on economic policy uncertainty that dies off after 5 months. Similarly, our results for the responses of economic policy uncertainty to shocks in the real interest rate and industrial production are consistent with our nonparametric estimates, lending support to the robustness of our main nonparametric results.

In sum, our parametric estimates reported in Fig. 6 generally support our results from the nonparametric estimations in the sense that they are suggestive that oil price shocks had a negative effect on economic policy uncertainty in the pre-GFC period and a strong positive and significant effect in the post-GFC period. It is worth noting that the parametric estimates provide only a summary of the average relationship between the variables. In this sense, the nonparametric estimates outperform the parametric counterparts by providing a complete picture of the entire distribution of economic

policy uncertainty conditional on oil price variables. In sum, our main nonparametric results are robust.

5. Concluding remarks

We have examined the effect of oil price volatility on economic policy uncertainty in the G7 countries using a time-varying nonparametric panel data model. Specifically, we employed a data-driven local linear method to empirically examine the time-varying trend and coefficient functions that govern the relationship between the oil price volatility and economic policy uncertainty. We allow these functions to evolve over time in the form of unknown functional forms with confidence bands constructed using a bootstrapping method.

Our approach has several advantages over parametric models used in the extant literature to model the relationship between the oil price and economic policy uncertainty. First, unlike the commonly adopted restrictive parametric specifications that have the risk of misspecification, our nonparametric panel data technique does not impose assumptions on the functional form. Instead, we allow the data to tailor the shape of the functional form. Second, we allow the effect of oil price volatility on economic policy uncertainty to evolve over time as an unknown functional form. We also allow the common global trend to evolve over time as an unknown functional form. Third, we estimate the country specific trend functions to evaluate the patterns of individual trend functions against the common global trend function.

Using monthly data for the period January 1997 to June 2018, our LLDVE estimates reveal that the relationship between oil price and economic policy uncertainty is time-varying. The coefficient function of the oil price was largely negative and significant for the period before the GFC and briefly negative at the end of the sample period. The coefficient function was positive and strongly significant following the GFC, except for the decoupling relationship at the end of the study period. We attribute this finding to the main driver of oil prices in recent decades before GFC being the surge in aggregate demand which has had a positive effect on the economic outlook and reduced policy uncertainty. We show that the country-specific trend functions follow the global common trend function for most countries. Our results reveal that the nonparametric estimates capture the complex nonlinear and time-varying relationship between

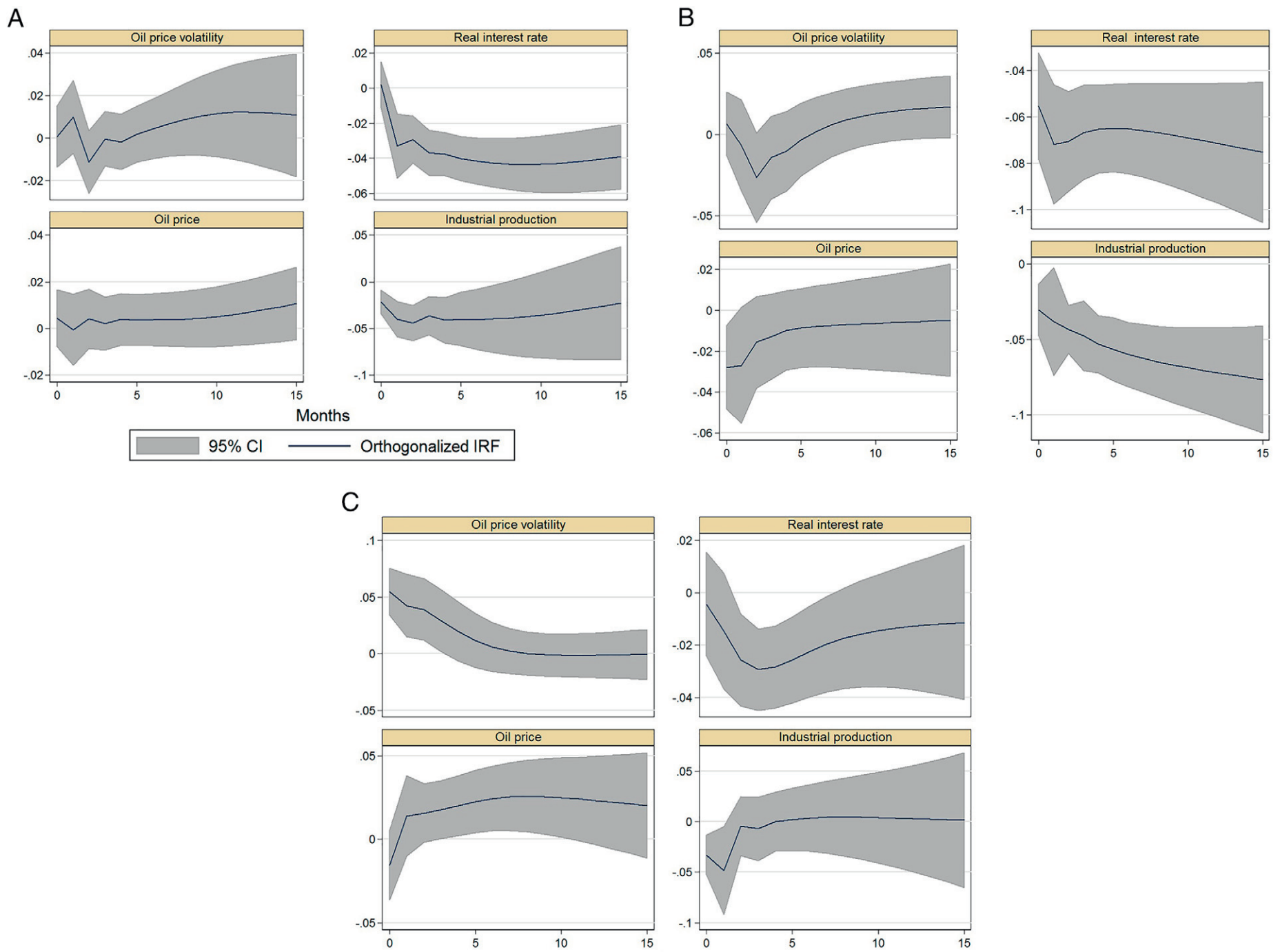


Fig. 6. The responses of economic policy uncertainty to the various shocks.

oil prices and economic policy uncertainty by accounting for the entire distribution of economic policy uncertainty.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eneco.2019.06.010>.

References

- Alquist, R., Kilian, L., Vigfusson, R.J., 2013. Forecasting the price of oil. *Handbook of Economic Forecasting*, vol. 2. Elsevier, pp. 427–507.
- Andrews, D.W., Lu, B., 2001. Consistent model and moment selection procedures for GMM estimation with application to dynamic panel data models. *J. Econ.* 101 (1), 123–164.
- Antonakakis, N., Chatziantoniou, I., Filis, G., 2014. Dynamic spillovers of oil price shocks and economic policy uncertainty. *Energy Econ.* 44, 433–447.
- Baker, S.R., Bloom, N., Davis, S.J., 2016. Measuring economic policy uncertainty. *Q. J. Econ.* 131 (4), 1593–1636.
- Barrero, J.M., Bloom, N., Wright, I., 2017. Short and long run uncertainty (no. w23676). *Natl. Bur. Econ. Res.*
- Bataa, E., Park, C., 2017. Is the recent low oil price attributable to the shale revolution? *Energy Econ.* 67, 72–82.
- Berger, T., Grabert, S., Kempa, B., 2016. Global and country-specific output growth uncertainty and macroeconomic performance. *Oxf. Bull. Econ. Stat.* 78 (5), 694–716.
- Blanchard, O.J., Gali, J., 2007. The macroeconomic effects of oil shocks: why are the 2000s so different from the 1970s? (no. w13368). *Natl. Bur. Econ. Res.*
- Blanchard, O.J., Riggi, M., 2013. Why are the 2000s so different from the 1970s? A structural interpretation of changes in the macroeconomic effects of oil prices. *J. Eur. Econ. Assoc.* 11 (5), 1032–1052.
- Bloom, N., 2009. The impact of uncertainty shocks. *Econometrica* 77 (3), 623–685.
- Bloom, N., 2014. Fluctuations in uncertainty. *J. Econ. Perspect.* 28 (2), 153–176.
- Bordo, M.D., Duca, J.V., Koch, C., 2016. Economic policy uncertainty and the credit channel: aggregate and bank level US evidence over several decades. *J. Financ. Stab.* 26, 90–106.
- Brogaard, J., Detzel, A., 2015. The asset-pricing implications of government economic policy uncertainty. *Manag. Sci.* 61 (1), 3–18.
- Caggiano, G., Castelnuovo, E., Groshenny, N., 2014. Uncertainty shocks and unemployment dynamics in US recessions. *J. Monet. Econ.* 67, 78–92.
- Caggiano, G., Castelnuovo, E., Figueres, J.M., 2017. Economic policy uncertainty and unemployment in the United States: a nonlinear approach. *Econ. Lett.* 151, 31–34.
- Christoffersen, P., Pan, X.N., 2018. Oil volatility risk and expected stock returns. *J. Bank. Financ.* 95, 5–26.
- Chudik, A., Pesaran, M.H., 2015. Common correlated effects estimation of heterogeneous dynamic panel data models with weakly exogenous regressors. *J. Econ.* 188 (2), 393–420.
- Cognigni, A., Manera, M., 2008. Oil prices, inflation and interest rates in a structural cointegrated VAR model for the G-7 countries. *Energy Econ.* 30 (3), 856–888.
- Degiannakis, S., Filis, G., Panagiotakopoulou, S., 2018. Oil price shocks and uncertainty: how stable is their relationship over time? *Econ. Model.* 72, 42–53.
- Duca, J.V., Saving, J.L., 2018. What drives economic policy uncertainty in the long and short runs: European and US evidence over several decades. *J. Macroecon.* 55, 128–145.
- Feng, J., Wang, Y., Yin, L., 2017. Oil volatility risk and stock market volatility predictability: evidence from G7 countries. *Energy Econ.* 68, 240–254.
- Filis, G., 2010. Macro economy, stock market and oil prices: do meaningful relationships exist among their cyclical fluctuations? *Energy Econ.* 32 (4), 877–886.
- Hamilton, J.D., 1983. Oil and the macroeconomy since World War II. *J. Polit. Econ.* 91 (2), 228–248.
- Hamilton, J.D., 2003. What is an oil shock? *J. Econ.* 113 (2), 363–398.
- D.J., Henderson, C.F., Parmeter, 2015. *Applied nonparametric econometrics*, Cambridge University Press
- Henderson, D.J., Kumbhakar, S.C., Parmeter, C.F., 2012. A simple method to visualize results in nonlinear regression models. *Econ. Lett.* 117 (3), 578–581.
- Henderson, D.J., Papageorgiou, C., Parmeter, C.F., 2013. Who benefits from financial development? New methods, new evidence. *Eur. Econ. Rev.* 63, 47–67.
- Kang, W., Ratti, R.A., 2013a. Oil shocks, policy uncertainty and stock market return. *Journal of International Financial Markets, Inst. Money* 26, 305–318.
- Kang, W., Ratti, R.A., 2013b. Structural oil price shocks and policy uncertainty. *Econ. Model.* 35, 314–319.
- Kang, W., Ratti, R.A., Vespignani, J.L., 2017. Oil price shocks and policy uncertainty: new evidence on the effects of US and non-US oil production. *Energy Econ.* 66, 536–546.
- Kendall, M.G., 1975. *Rank Correlation Methods*. Griffin, London.
- Kilian, L., 2008. Exogenous oil supply shocks: how big are they and how much do they matter for the US economy? *Rev. Econ. Stat.* 90 (2), 216–240.
- Kilian, L., 2009. Not all oil price shocks are alike: disentangling demand and supply shocks in the crude oil market. *Am. Econ. Rev.* 99 (3), 1053–1069.
- Kilian, L., 2017. The impact of the fracking boom on Arab oil producers. *Energy J.* 38 (6), 137–160.
- Kilian, L., Park, C., 2009. The impact of oil price shocks on the US stock market. *Int. Econ. Rev.* 50 (4), 1267–1287.
- Li, D., Chen, J., Gao, J., 2011. Nonparametric time-varying coefficient panel data models with fixed effects. *Econ. J.* 14 (3), 387–408.
- Lippi, F., Nobili, A., 2012. Oil and the macroeconomy: a quantitative structural analysis. *J. Eur. Econ. Assoc.* 10 (5), 1059–1083.
- Mammen, E., 1993. Bootstrap and wild bootstrap for high dimensional linear models. *Ann. Stat.* 255–285.
- Mann, H.B., 1945. Nonparametric tests against trend. *Econometrica* 13 (3), 245–259.
- Middleton, R.S., Gupta, R., Hyman, J.D., Viswanathan, H.S., 2017. The shale gas revolution: barriers, sustainability, and emerging opportunities. *Appl. Energy* 199, 88–95.
- Mumtaz, H., Theodoridis, K., 2017. Common and country specific economic uncertainty. *J. Int. Econ.* 105, 205–216.
- Peersman, G., Van Robays, I., 2012. Cross-country differences in the effects of oil shocks. *Energy Econ.* 34 (5), 1532–1547.
- Pesaran, M.H., 2004. General diagnostic tests for cross section dependence in panels. *Cambridge Working Papers in Economics No. 0435*. University of Cambridge.
- Pesaran, M.H., Shin, Y., Smith, R.P., 1999. Pooled mean group estimation of dynamic heterogeneous panels. *J. Am. Stat. Assoc.* 94 (446), 621–634.
- Ratti, R.A., Vespignani, J.L., 2016. Oil prices and global factor macroeconomic variables. *Energy Econ.* 59, 198–212.
- Rehman, M.U., 2018. Do oil shocks predict economic policy uncertainty? *Phys. A* 498, 123–136.
- Silvapulle, P., Smyth, R., Zhang, X., Fenech, J.P., 2017. Nonparametric panel data model for crude oil and stock market prices in net oil importing countries. *Energy Econ.* 67, 255–267.
- Su, L., Ullah, A., 2008. Local polynomial estimation of nonparametric simultaneous equations models. *J. Econ.* 144 (1), 193–218.
- Sun, Y., Carroll, R.J., Li, D., 2009. Semiparametric estimation of fixed-effects panel data varying coefficient models. *Nonparametric Econometric Methods*. Emerald, pp. 101–129.
- Tang, W., Wu, L., Zhang, Z., 2010. Oil price shocks and their short-and long-term effects on the Chinese economy. *Energy Econ.* 32, S3–S14.
- You, W., Guo, Y., Zhu, H., Tang, Y., 2017. Oil price shocks, economic policy uncertainty and industry stock returns in China: asymmetric effects with quantile regression. *Energy Econ.* 68, 1–18.